

Exploring Interactive Gestures with Voice Assistant on HMDs in Social Situations

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ABSTRACT

Voice assistants (VAs) are growing in popularity within VR and AR. However, speaking to the VA publicly can lead to misunderstanding and awkwardness. In this paper, we explored the interactive technique that enabled users to indicate whether to talk to the VA in a sociable manner. We initially collected 62 switching gestures leveraging various body parts (e.g., head and hand) through a brainstorming session with ten experts, then voted on the best ones. Our findings emphasized the speak-to-shoulder gesture. We then developed the detection system and assessed its performance using real data collected from 15 participants. The results showed an accuracy of 98.0%, and a false triggering rate of 0.10 times per minute.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Sound-based input / output

1 INTRODUCTION

With the rapid development of speech recognition algorithms, voice assistants (VAs) are becoming increasingly popular in our daily lives. The ability of eyes-free and hands-free interaction makes interacting with VAs superior to typical touch interaction in smart IoT scenarios. Moreover, voice interaction has also become an attractive solution for virtual reality (VR) and augmented reality (AR).

However, interacting with VAs in social situations may lead to misunderstanding, as the people around the user may not be able to distinguish the object of the voice command. The majority of VAs today used keyword spotting (KWS) to issue voice commands (e.g., “Hey Siri”) and prevent such confusion. However, this would considerably lower interaction effectiveness and potentially cause awkward situations when used in social settings.

In this work, we explored how to interact with VAs in a sociable manner, with the goal of achieving a balance between interaction performance and social experience. Specifically, we chose head-mounted-displays (HMDs) as the apparatus, as we envisioned the pervasiveness of social VR [2] and mobile AR [1].

To elicit the possible gestures, we conducted a brainstorming session with 10 experts and generated a total of 62 gestures that

leveraged body movements or acoustic features. Then, we conducted another user study to assess the six top voted gestures from the perspective of both the speaker and the listener. Our research underscored the speak-to-shoulder gesture as the most favored.

Based on the findings, we proposed our interactive technique for keyword-free voice interaction with VAs in society. The user could perform the gesture above to indicate the intention of briefly speaking with the VA, which is also easy for the listener to understand. After the interaction, the users can turn their heads back to neutral orientation, and continue talking to the listener.

We designed and validated the algorithm based on the collected data from 15 users, including both positive samples when performing the gesture, and negative samples during wandering and chatting. Simulation results indicated our system achieved an accuracy of 98.00%, and a false triggering rate of 0.10 times per minute.

2 STUDY 1: BRAINSTORMING THE VOICE INPUT SWITCHING GESTURE IN SOCIAL SETTINGS

The session is designed to enable participants to design and discuss collaboratively, guiding consensus toward high-quality results. This session engaged 10 knowledgeable participants from our campus specializing in VR interaction. The group comprised 8 males and 2 females, aged between 19 and 26. All participants were educational or academic professionals; 8 had experience in VR/AR research, and 2 had experience using VR/AR devices. Each participant had prior experience in conducting interactive research on smart devices.

We sought to explore the ability of users to seamlessly switch interaction targets between third parties and VAs in VR/AR environments. To maintain gesture practicality, we imposed one limitation: users should not directly utilize facial expressions, as their excessive use may interfere with social interactions.

We commenced by assembling participants in a quiet conference room, allowing them some time to familiarize themselves with each other. Initially, each participant was given 15 minutes to independently formulate ideas. Subsequently, participants sequentially presented their ideas, affixing them to the board. Following the presentation, a group discussion was held to address any overlooked ideas. Ultimately, participants were requested to cast their votes for up to 20 of the most satisfactory ideas. Ideas garnering votes were considered of superior quality. Voting was guided by the following criteria: confusion, simplicity, fatigue, and embarrassment.

2.1 Results

The users proposed a total of 62 valid ideas, as summarized in Table 1. The gestures were classified based on the action parts involved and the types of signs used. The action parts were determined by the body parts engaged in executing the motions, including 4 parts—*foot*, *hand*, *head* and *mouth*. Due to gesture combinations, several cases integrated multiple movement parts. The gestures were segregated into two categories based on their sign type: *motion* and *acoustic*. The *motion* category involved one or more complete actions serving as the sign, as seen in “touch the HMD”. Conversely,

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the *acoustic* type employed voice as trigger signs, exemplified by “snap fingers”. According to Table 1, gestures combining *hand* and *motion* were predominant (46 out of 62), indicating that users found hand movements to be the most significant and familiar method.

Table 1: Statistics of all the 62 brainstormed gestures. One gesture leveraged “hand”+“head”, thus was counted twice.

Form \ Part	Foot	Hand	Head	Mouth
Motion	4	46	5	1
Acoustic	4	2	0	1
Total	8	48	5	2

Among 10 voters, each of them has cast 7 to 13 votes (mean: 10.0, std: 1.76). The numbers suggest that each voter would focus on ideas that are qualified. To evaluate the reliability of voting, we calculated the intra-class correlation [3] of the result. The corresponding indicator for this experiment was $ICC(3, k)$, meaning “reliability of the mean rating” in Case 3, and the result yielded $ICC(3, k) = 0.811$. Thus we finally took the ideas with a mean rating over a half (i.e. approximate to one) as the passed ideas.

The finally selected six gestures were depicted in Figure 1. G1 (8 votes) and G2 (8 votes) indicated the intent to interact with the HMD. Conversely, G3 (7 votes) and G4 (6 votes) signified the intention to cover the mouth. G5 (6 votes) utilized both hands instead of the head, showing an intent to wake up the HMD. Lastly, G6 (6 votes) only required a user’s head movement, imitating the action of speaking to an imagined widget on the collar.

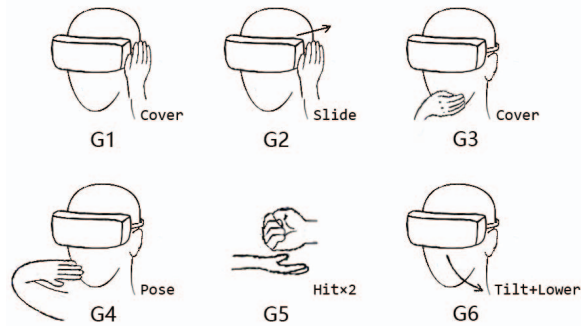


Figure 1: The six gestures that received over a half votes.

3 IMPLEMENTATION OF TECHNIQUE

The basic concept of this technique was to shift the head to switch between conversations of human and VA. Similar concepts that utilize head-based gestures have proven effective, such as HeadGesture [4]. Nevertheless, our work was beyond previous efforts because we addressed the conversation switching between people and VA in a more socially acceptable way.

The system workflow is outlined as follows. Initially, the HMD sensors capture the user’s head movements and voice through its integrated IMU and microphone. Within the HMD, we set up an RNN model to determine if a user gesture has been triggered, relaying this information back to the system. If triggered, speech data from users would be sent to the voice assistant.

3.1 User Data Collection

We recruited 9 males and 6 females from the campus to collect user’s data. A PICO 4 was used as the HMD. We collected both pos-

itive and negative data with several primary tasks. Each participant completed these tasks in the order derived from Latin squares.

For negative scenario, two primary tasks were designed to collect natural head behaviors. One task let users freely walk and look around in a room, the other asked them to sit down and chat with the experimenter. Both tasks lasted for 5 minutes. Ultimately, we collected 150-minute natural data with 75 for either primary tasks.

For positive scenario, one primary task was designed that required users to actively perform gestures. The process incorporated a warm-up exercise. The actual task comprised 30 conditions as sub-tasks made up of 10 voice commands and 3 initial head orientations—up, down, and forward. For each sub-task, users began by pressing the button, performed the required action, then pressed the button again to conclude. During data processing, we tracked the action sequence from the origin and discarded any preceding data. With 15 users and 30 conditions in total, we gathered a collection of 450 records.

3.2 Performance Simulation

We evaluated the performance of our system on the collected data. The 450 positive gesture cases are truncated at the motion extreme, eliminating any subsequent recovery actions. This setup is designed to verify that the system’s activation relies exclusively on trigger actions and not on recovery actions. The total 150-minute negative data is divided into 900 segments of 10 seconds each.

According to the result, our RNN’s accuracy is 98%. This indicates that our algorithm maintains stable recognition capabilities and its effectiveness is minimally impacted by other conditions. We also examined the false triggering rate of our system on the negative data, yielding 0.10 times per minute.

4 CONCLUSION

Voice assistants are becoming increasingly prevalent in VR or AR. However, with the increasing number of users in VR or AR, speaking to the VA in social scenarios may lead to misunderstanding and embarrassment. Aiming at triggering the VA in a more efficient and natural way, we proposed this technique that leveraged speak-to-shoulder gestures for users to switch conversational targets to VA. Through a brainstorming with 10 experts, we first collected a total of 62 sociable interactive gestures with VAs that leveraged different body parts (e.g., head and hand), reaching 6 candidate gestures through voting. The speak-to-shoulder gesture was selected according to further study. Based on the collected data from 15 real users, our system reached an accuracy of 98.0%, and a false triggering rate of 0.10 times per minute.

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